

Assignment 4

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1 Monte-Carlo Simulation

SYSTEM:"N = 108, rho = 0.8442, Temp = 0.728, max-del = 0.2"

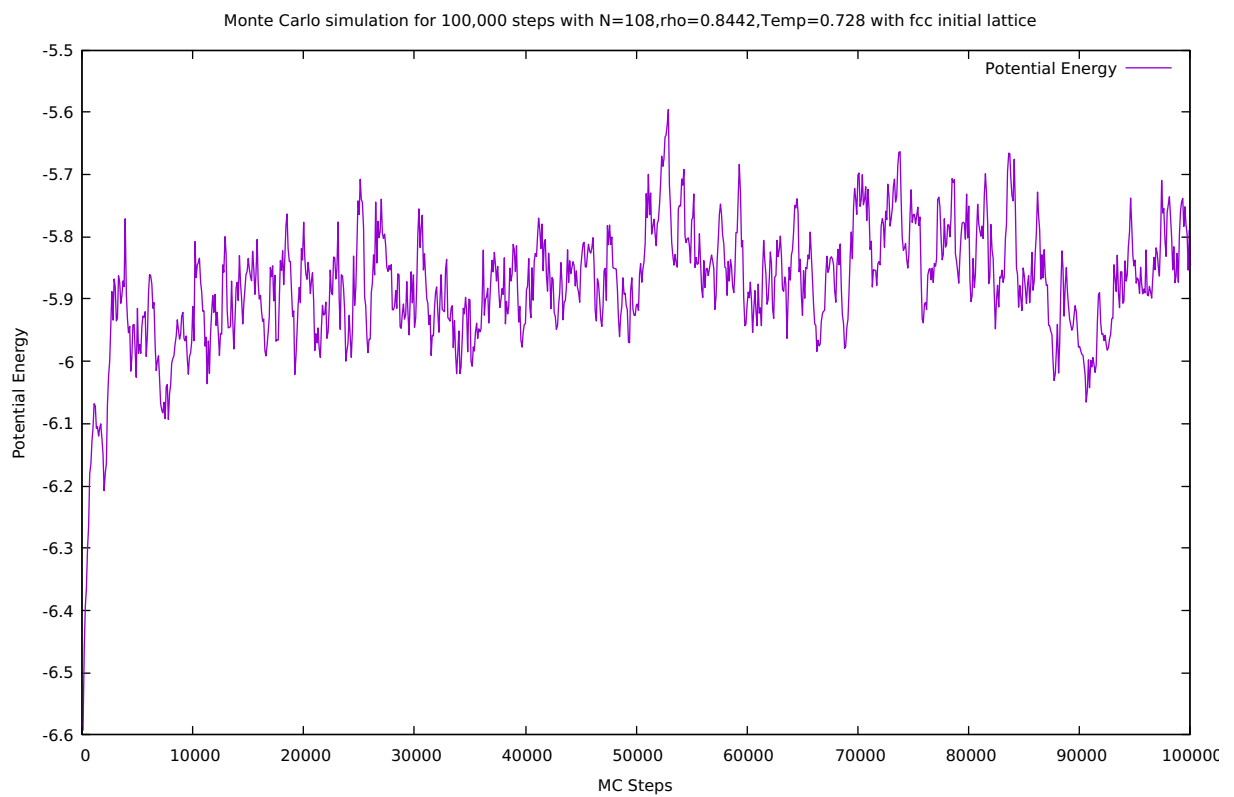


Figure 1: Monte Carlo simulation for 100,000 steps with N=108,rho=0.8442,Temp=0.728 with fcc initial lattice

1.1 Output

Initial Potential Energy: -5.6932774238161770

Final Potential Energy: -4.8879520968559014

The number of accepted move 46603.000000000000

The acceptance probability: 46.603000000000002

Program run time : 0.9848349999999990

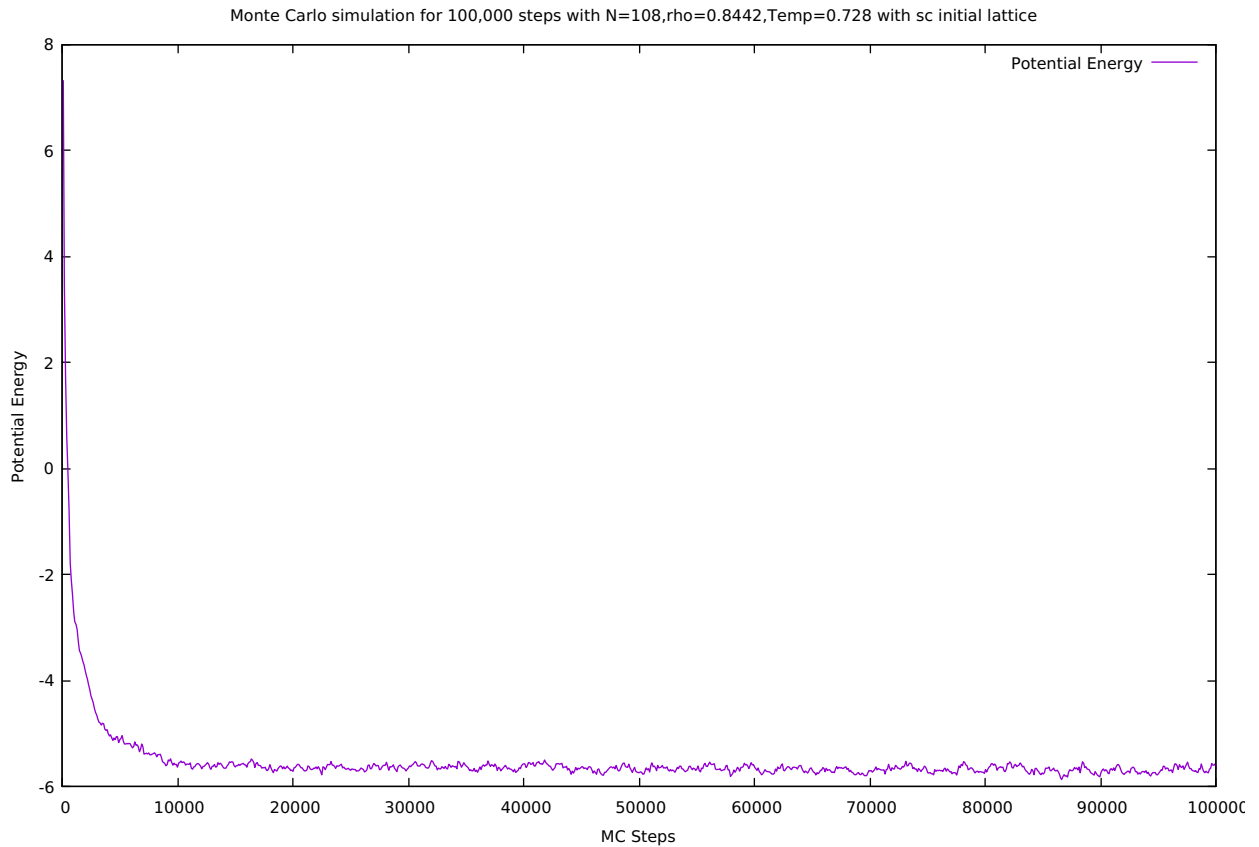


Figure 2: Monte Carlo simulation for 100,000 steps with $N=108$, $\rho=0.8442$, $\text{Temp}=0.728$ with sc initial lattice

1.2 Output

Initial Potential Energy: 13.329583343241142

Final Potential Energy: -4.4419750929519388

The number of accepted move 41362.000000000000

The acceptance probability: 41.362000000000002

Program run time : 0.9994849999999996

2 Molecular Dynamic Simulation

" $N = 108$, $\rho = 0.8442$, $\text{Temp} = 0.728$, $\text{steps}=100,000$ ($t_f=100$, $dt=0.001$)"

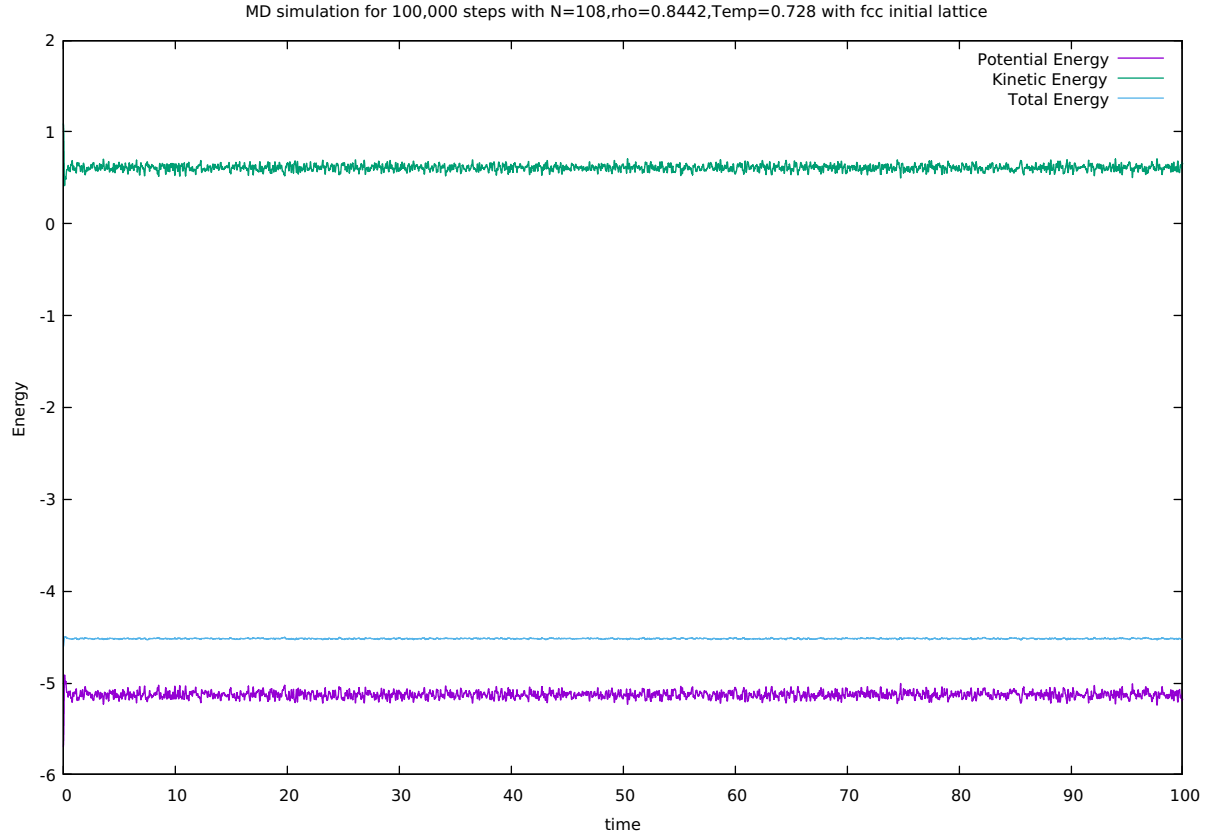


Figure 3: MD simulation for 100,000 steps with $N=108$, $\rho=0.8442$, $Temp=0.728$ with fcc initial lattice

2.1 Output

scaling factor(fs): 3.0045976998167458
 Initial Temperature: 0.72024869477706566
 Initial Potential Energy: -5.6932774238161770
 Initial Kinetic Energy: 1.0803730421655986
 Final Temperature: 0.38825391088198613
 Final Potential Energy: -5.1074657074132750
 Final Kinetic Energy: 0.58238086632297925
 Program run time : 24.333801000000001

3 Comparison between MC vs MD

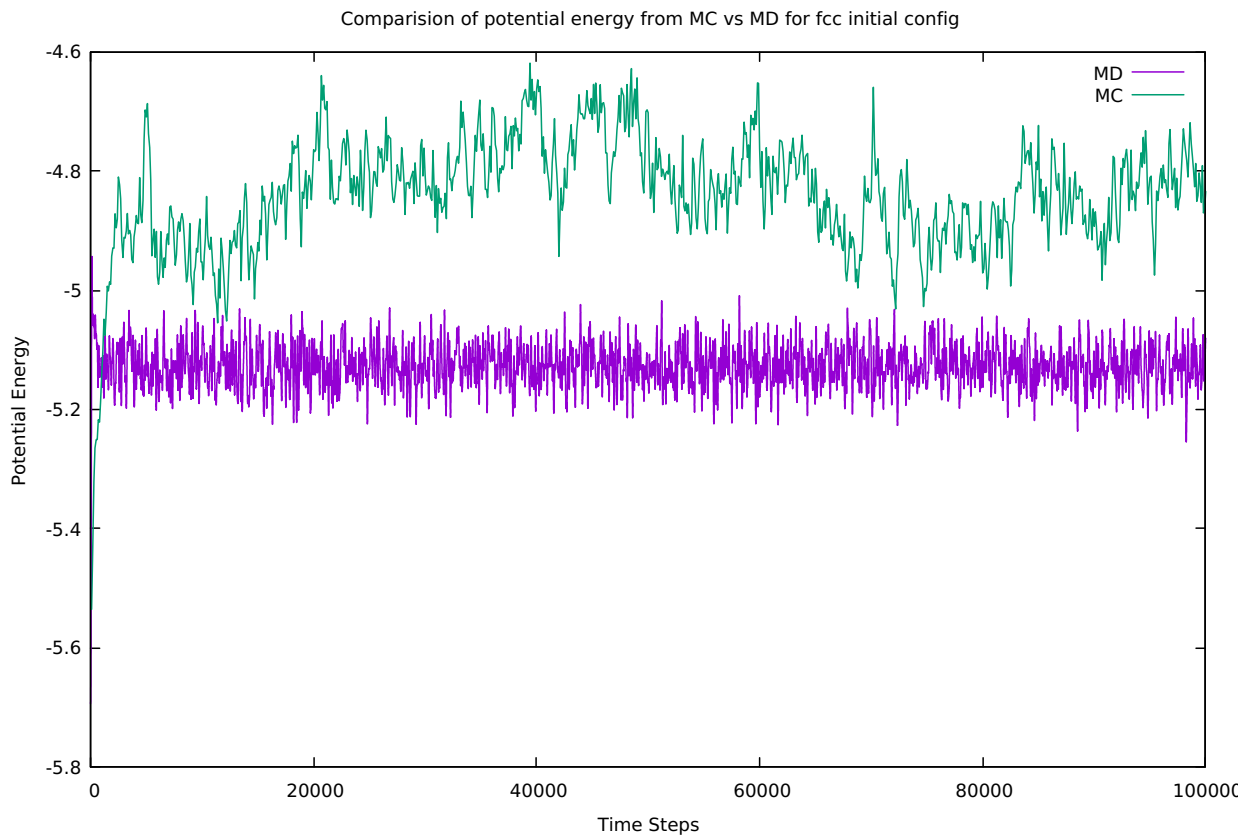


Figure 4: comparison between MC and MD simulation